Week 39: Road detours

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This view repeats, by problem, the answer keys of the print guide, week-39-facilitator.pdf, whose source is editable/facilitator-src/content.json and route-note.json; page references in the keys are to that guide. Change both together; check_guide_copy.py compares them.

Mathematical overview

Mathematical purpose

A journey is an ordered finite list of directed traversals of actual edges on a fixed graph. The only shortening move deletes two adjacent traversals of the same edge in opposite directions. The graph, start, and finish remain fixed. The empty journey means staying at the initial vertex.

Unique reduction. Every journey ends in the same reduced edge path, regardless of deletion order. Each deletion shortens by two, so the process terminates. A running stack of oriented edges gives an invariant normal form; the proof is on page 6. “Same endpoints” is enough to identify an edge only on these simple printed graphs. Parallel edges would need separate road identities.

Tree theorem. Between two vertices of a tree there is a unique simple path, and every journey between them reduces to it. In particular, every closed tree journey reduces to the empty journey. A reduced path on a four-edge ring, by contrast, keeps going in a single direction and may survive for one or more complete circuits.

Order can survive. Let a be H→A→B→H and b be H→C→D→H on the two-ring map. The journeys ab and ba have distinct reduced paths. The commutator ab a^{-1} b^{-1} has zero net count around each ring but has no adjacent inverse edges, so it does not disappear. Net traversal counts lose essential order information.

Topological scope. These elementary edge moves describe endpoint-fixed homotopy of graph edge paths, with immediate inverse-pair insertions allowed when comparing equivalence classes. The student shortening activity uses deletions. The open regions inside rings are not filled faces: paths cannot slide across them, cancel separated inverse steps, trade subjourney order, or move their basepoint. Paths may revisit vertices and overlap themselves.

Entry points and readiness

K-1: play and replay short journeys with a pawn; an adult records ordered steps. **Grades 2-3:** construct long tree walks, classify ring results, and contrast order on two rings. **Grades 4-5:** prove unique reduction and find a zero-net-count trip that cannot disappear. Counting to 12 is useful, not fluent reading or algebra.

Before-use gate: the exact large route tiles, adult-recording method, mats, and pawn workflow have not been physically tried with the intended group. A K-1 child must be able to replay an ordered route and remove an immediate reversal without changing the remaining order. Stay with the concrete demonstration if that is not yet reliable.

Reading the keys

In this key a compact route such as ABDE means A→B→D→E. Count edges, not letters: a seven-step route has eight displayed vertices. The starting letter remains when every step is removed.

Problems and routes

Problem	K–1	2–3	4–5	route
Shortening trips on the tree	P1	P1	P1	first
Four-step trips home on the ring	P2			first
Seven-step trips on the tree	P3			return
Trips home on the tree	P4	P2	P2	K-1 return; 2-3 first; 4-5 first
Trips from C to F	P5		P3	K-1 return; 4-5 first
What eight-step ring trips leave	P6	P3		K-1 return; 2-3 first
Does the removal order matter?		P4	P4	first
What six-step trips from A to C leave	P7			return
Two rings in either order		P5	P5	return
Same ring counts, different results		P6	P6	return

First route. K–1 P1–P2; Grades 2–3 P1–P4; Grades 4–5 P1–P4. **Return visits.** K–1 P3–P7; Grades 2–3 P5–P6; Grades 4–5 P5–P6.

Key, one entry per problem

Shortening trips on the tree

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. On a tree every trip shortens to the one simple path between its start and finish, whichever out-and-back pairs are removed first.

K–1: The three-step A-to-E trip is A-B-D-E. Every four-step B-home trip reduces to staying at B; every four-step A-to-D trip reduces to A-B-D. For example B-A-B-A-B disappears, while A-B-A-B-D leaves A-B-D.

Grades 2–3: The three printed routes reduce, in order, to ABDE, ABDE, and CBDF. One different seven-step route from C to F is CBABDEDF. Its two removable excursions are B→A→B and D→E→D; what remains is CBDF.

Grades 4–5: Printed results: ABDE, ABDE, CBDF. The seven-step construction CBABDEDF is one valid new C-to-F input with reduced path CBDF. The listed C-to-F input has five steps; the new route must meet the requested seven-step constraint.

Four-step trips home on the ring

K–1 P2 · first route

Idea. A four-step trip home on the four-road ring either shortens to staying at A or keeps all four steps, one circuit in either direction.

The possible reduced four-step ring home results are stay at A, A-B-C-D-A, and A-D-C-B-A. A nonempty reduced ring trip keeps one direction; the two circuits survive.

Seven-step trips on the tree

K–1 P3 · return route

Idea. Every seven-step trip from A to E shortens to A→B→D→E, and every one from C to F to C→B→D→F: the simple paths in the tree.

Example seven-step A-to-E trip: ABABDFDE, reducing to ABDE. Example seven-step C-to-F trip: CBABDEDF, reducing to CBDF. Any valid constructions have these same respective reduced routes, the unique simple paths in the tree.

Trips home on the tree

K–1 P4 · Grades 2–3 P2 · Grades 4–5 P2 · K-1 return; 2-3 first; 4-5 first route · wording differs by band

Idea. Every trip from A back to A on a tree shortens to staying at A: a farthest place visited is always left by the road just used.

K–1: All 4-, 6-, 8-, and 12-step trips from A back to A shorten to staying at A. Explicit witnesses alternate A and B for 2, 3, 4, or 6 out-and-back pairs. Encourage less repetitive excursions when children are ready; the theorem covers all valid choices.

Grades 2–3: Valid 12-step A-home trips covering every road include ABABCBDEDFDBA, ABABCBDFDEDBA, and ABABDEDFDBCBA. Every one reduces to A. A nonempty reduced closed walk in a tree would force a cycle; equivalently a farthest excursion forces a reversal. The claim holds for all finite lengths, not just these examples.

Grades 4–5: No nonempty A-home trip on the tree survives. Root the tree at A. In any nonempty closed journey, take a visited vertex of greatest distance from A. Choose an occurrence reached from its parent. Since no deeper vertex is visited and a tree gives it only one neighbor closer to A, the next step returns along that same edge. There is a cancellable pair. Reapply after each deletion until only A remains.

Trips from C to F

K–1 P5 · Grades 4–5 P3 · K-1 return; 4-5 first route · wording differs by band

Idea. Every trip from C to F, however long, shortens to C→B→D→F, so no extra step survives and no two trips finish differently.

K–1: CBABDEDBDF has nine steps, uses AB, BC, BD, DE, and DF, and ends at F. Delete B→A→B, D→E→D, and the later D→B→D excursion as they become adjacent to get CBDF. No two can end differently: every trip from C to F shortens to C→B→D→F.

Grades 4–5: No. Every C-to-F journey reduces to CBDF. A reduced tree path cannot revisit a vertex, because the subjourney between repeated visits would be a nonempty closed reduced tree walk. It is thus simple, and a tree has only one simple path between the endpoints.

What eight-step ring trips leave

K–1 P6 · Grades 2–3 P3 · K-1 return; 2-3 first route · wording differs by band

Idea. Eight-step trips home on the ring leave exactly five results: staying at A, or one or two circuits in either direction.

K–1: There are exactly five reduced home results from eight-step ring walks:

Remaining route	Length	One eight-step input
Stay at A	0	ABABABABA
ABCDA	4	ABABABCDA
ADCBA	4	ABABADCBA
ABCDABCDA	8	ABCDABCDA
ADCBADCBA	8	ADCBADCBA

Completeness: on a degree-two ring, once a nonempty reduced route chooses a direction it cannot turn around. A home route has length a multiple of four. Deletions preserve parity and cannot increase length; among lengths 0 through 8 this gives 0, 4, or 8, with two directions for each nonzero length. The listed inputs prove attainability.

Grades 2–3: The complete eight-step ring result set is A (stay), ABCDA, ADCBA, ABCDABCDA, and ADCBADCBA. The fewest steps in a nonempty surviving trip home is four. A degree-two ring leaves a reduced walker no choice except continuing in its chosen direction; a closed route must complete full four-edge circuits. The construction table on page 3 provides eight-step witnesses for every result.

Does the removal order matter?

Grades 2–3 P4 · Grades 4–5 P4 · first route · wording differs by band

Idea. The final route does not depend on which reversal is removed first, on any road map; a left-to-right stack proves it.

Grades 2–3: ABABCBADCDA always reduces to A. One order is:
ABABCBADCDA → ABCBADCDA → ABADCDA → ADCDA → ADA → A.
Another begins by removing the B→C→B excursion:
ABABCBADCDA → ABABADCDA → ABADCDA → ADCDA → ADA → A.
Each arrow deletes one adjacent same-road reversal. Different intermediate routes do not imply different final routes.

Grades 4–5: The three printed routes reduce to A, ADC, and ABCDA, respectively. The page 6 proof shows no choice of first reversal can change the final reduced edge path on any graph. Illustrative reductions for the last two:
ABADCBC → ADCBC → ADC.
ABCBCDABA → ABCDABA → ABCDA, or ABCBCDABA → ABCBCDA → ABCDA.
The third route has three possible first removals and four removal orders, all ending at ABCDA.
For the first, see the two distinct deletion orders on page 4. These computations illustrate the theorem; they are not its proof.

What six-step trips from A to C leave

K–1 P7 · return route

Idea. Six-step trips from A to C on the ring leave exactly four results: ABC, ADC, ABCDABC and ADCBADC.

Exactly four results: ABC, ADC, ABCDABC, and ADCBADC. Their lengths are 2, 2, 6, and 6. Six-step inputs for the short results include ABABABC and ABABADC. A nonreversing route from A to the opposite vertex C has length 2 modulo 4, so no other result of length at most 6 exists. Yes, the two six-step survivors both use all four roads, but in different orders. They do not have the same directed-edge multiset; the question only asks for the same roads. Accept no with the reason that the tiles differ.

Two rings in either order

Grades 2–3 P5 · Grades 4–5 P5 · return route · wording differs by band

Idea. On two rings joined at H, the order of whole ring trips matters: the same ring trips in another order can shorten to a different trip, and left, right, left backward, right backward cannot disappear.

Grades 2–3: Write a for left forward (HABH), a^{-1} for left backward (HBAH) and b for right forward (HCDH). Four orders shorten to HCDH: a a^{-1} b, a^{-1} a b, b a a^{-1} and b a^{-1} a, because a ring trip next to its backward trip cancels. The other two, a b a^{-1} = HABHCDHBAH and a^{-1} b a = HBAHCDHABH, contain no out-and-back pair, so they stay as they are and differ. The same three ring trips in a different order can give a different result.

Grades 4–5: The requested journey is HABHCDHBAHDCH, or ab a^{-1} b^{-1}. No consecutive pair is an edge followed immediately by that same edge backward, including at H between circuits. Nothing can be removed, so the trip cannot disappear. H is the based endpoint and interior junction; it is not permission to split off or reorder a circuit.

Same ring counts, different results

Grades 2–3 P6 · Grades 4–5 P6 · return route · wording differs by band

Idea. Going around each ring once each way does not decide whether a trip disappears; the order of the circuits does.

Grades 2–3: A 12-step disappearing example is HABHBAHCDHDCH: left forward, left backward, right forward, right backward. A 12-step survivor is HABHCDHBAHDCH: left forward, right forward, left backward, right backward. Both use both rings. In the first, nested opposite traversals cancel within each round-trip pair. The second has no immediate reversal anywhere.

Grades 4–5: Of the 24 orders of a, b, a^{-1}, b^{-1}, 16 disappear and 8 survive. If two trips on the same ring are next to each other, they are a ring trip and its backward trip, which cancel; the other ring's two trips are then next to each other and cancel too. If the rings alternate, left, right, left, right or right, left, right, left, no out-and-back pair occurs anywhere, including at H: these 8 orders survive (two patterns, each with two ways to place each ring's two trips), such as ab a^{-1} b^{-1}. Every order goes around each ring once each way, so equal net counts do not decide whether a trip disappears; the order does.

Optional hints and facilitation checks

Tree: "What happens when you arrive at a leaf?" Ring: "If you turn around now, did you create a removable pair?" Order: "Where is the first place the two journeys disagree?" Ask for an actual pawn replay to settle a claim that two steps use the same road. Do not cancel distant opposite traversals merely because their directions balance.

Source and verification

Allen Hatcher, *Algebraic Topology*, Chapter 1: §1.1 defines endpoint-fixed path homotopy; §1.2 treats reduced words/free products; §1A, "Graphs and Free Groups," develops the graph setting. The stack and finite ring arguments here are self-contained versions for the printed road maps. Chapter 1 PDF (`https://pi.math.cornell.edu/~hatcher/AT/ATch1.pdf`).